RoboCupJunior Rescue Simulation 2023

Team Description Paper

Fall'itis

Abstract

Our algorithm presents a significant improvement compared to existing systems by eliminating the reliance on the tile system. This unique feature empowers our robot to navigate smoothly through narrow passages and excel in the fourth zone. The key differentiating factor lies in the integration of a high-resolution map, enabling the robot to navigate efficiently in confined spaces. Additionally, our system seamlessly combines real-time object detection and pose detection, further enhancing its capabilities.

1. Introduction

Fall'itis is a team consisting of two members: Michele Tesser and Alin Odobescu.
Alin Odobescu has played a vital role in developing the mapping system, path planning, and navigation. His involvement in this competition has sparked an interest in machine learning and computer vision.
Michele Tesser has focused his efforts on object detection, dataset management, operating and scoring game elements, and analyzing various complex scenarios. His passion for mathematics has been instrumental in our success in this competition.
We take pride in our victory in the national phase of this competition. Michele has excelled in Italian statistics competitions, while Alin has achieved a national level recognition in computer science competitions. Both team members have actively participated in mathematics and computer science competitions prior to this event, creating a strong synergy and deepening their mutual understanding.

2. Project Planning

Our goal for the competition was to venture into new and unexplored fields such as robotics. It was our first time participating in a long-duration competition, which presented a unique opportunity for us to simulate a work environment. We had to plan our actions, perform project management tasks, and coordinate our team effectively.
Our project plan was focused on developing a robot capable of navigating seamlessly in the fourth zone, irrespective of the other zones. To achieve this, we utilized a LiDAR sensor to create an accurate map that could account for various obstacle configurations. Additionally, we implemented a real-time object detection model based on Ultralytics YOLOv8 to precisely position the signs on the map.
The milestones we have accomplished so far include:

- Developing an efficient LiDAR mapping algorithm using polar grid to cartesian grid conversion.
- Implementing an object detection system based on Ultralytics YOLOv8.

However, we have not yet achieved a satisfactory solution for path planning, which remains an ongoing focus of our work leading up to the competition.
Upon analyzing the competition rules, we noticed the introduction of a new zone, Zone 4, which captured our interest. Consequently, we dedicated efforts to develop an algorithm capable of exploring this zone, which eventually resulted in our robot being able to navigate the entire maze.
While observing other teams, we discovered that no one had considered moving away from the traditional tile system. However, this did not deter us from pursuing our goal.

3. Robot Design

a. LiDAR

One of the standout features of our robot is the integration of a LiDAR (Light Detection and Ranging) sensor. We specifically chose LiDAR due to its versatility and superior capabilities. With the ability to emit 2048 rays in 4 layers, covering a 360° field of view, the LiDAR provides us with a comprehensive environmental visualization. While there were cheaper distance sensors available, opting for LiDAR ensures a more effective understanding of the robot's surroundings by providing detailed information about the environment.

b. Color Sensor

Located on the robot's base, we have incorporated a color sensor to detect the color of the tile on which the robot is positioned. While computer vision was considered as an alternative, we found it to be excessive for our requirements. The color sensor emits a ray and returns a tensor that indicates the detected color, which is useful for creating a more detailed map. Although our general software implementation does not rely on knowing the specific zone we are in, the color sensor enhances our mapping capabilities.

c. GPS

To accurately determine the robot's location, we have integrated a GPS module. While odometry (calculating position based on wheel revolutions) could provide relative position information, certain scoring elements in the competition require knowledge of the robot's absolute position. Therefore, we opted for GPS, which provides a vector with the robot's three coordinates, ensuring precise localization.

d. Inertial Unit

Our robot is equipped with an inertial unit that extends beyond a simple gyroscope. It includes mathematical functions for converting angular velocities into angles, allowing us to accurately determine the robot's orientation. This capability enables us to create a global map instead of being limited to a local one. Odometry alone would not suffice, especially when encountering situations such as a Lack of Progress, where the robot needs to account for unknown orientations.

e. Distance Sensors

We have incorporated three distance sensors on the base of our robot for detecting black holes. While using only one color sensor would have been inadequate since the robot's movements are not confined to cardinal directions, these distance sensors measure the distance to the ground. They enable us to identify unexpected increases in distance, indicating the presence of a hole and allowing the robot to avoid it.

f. Cameras

Cameras play an indispensable role in our robot's functionality. They enable us to capture images of the maze and facilitate victim detection. We have strategically positioned two cameras at the right and left edges, aligned

with the center axis, to simplify mathematical calculations and provide a comprehensive view of the surroundings.

4. Software

a. General software architecture

Our software architecture is organized into the following sections:

- Motion: this component controls the robot's motion and updates the velocities of the wheel motors based on various inputs.
- Mapping: the mapping section is responsible for updating the map using LiDAR scans at each step of the Webots simulation. It also includes functionality for recognizing and locating signs within the environment. Additionally, it utilizes data from the distance sensors to detect and map holes, as well as the color sensor to identify and map special tiles.
- Navigation: the navigation component consists of our path planning algorithm, which determines the optimal path for the robot to follow. Additionally, it includes a correction algorithm that adjusts the path between map updates to account for any changes in the environment.
- Logic: the logic section houses the main decision-making processes of the robot. It determines when to trigger a "lack of progress" event and handles the robot's behavior in various situations.

By dividing our software into these distinct sections, we ensure clear separation of responsibilities and enable efficient coordination among different functional aspects of our robot's operation.

b. Navigation

Navigating the maze has posed a significant challenge for our team. Our path and motion planning algorithm is designed to effectively guide the robot through the maze by considering various inputs, such as mapping data, the robot's position, and yaw (orientation). The algorithm outputs the appropriate wheel speeds, ensuring smooth and controlled movements.
The navigation algorithm works as follows:

```
spots = walkable points adjacent to dilation zone
while not all spots are visited:
        nearest_point = find_nearest_unvisited_point()
        reach_point(nearest_point)
        while current_position not visited:
                follow_dilation_zone()
exit_maze()
```

To maintain a smooth trajectory, we adjust the wheel motor velocities based on the furthest of the nearby goal positions. This approach helps optimize the robot's movements and ensure efficient navigation.
Additionally, we have implemented a mechanism to handle situations where the path becomes invalid between map updates. In such cases, we utilize a breadth-first search (BFS) algorithm starting from the last valid path position to restore the path and ensure continuous navigation.
To ensure comprehensive exploration of the entire maze, we mark a region near the robot's position as "visited." We have chosen to mark a region instead of a single point due to the limitations in mapping and localization accuracy. The "visited" value expands over time, allowing us to determine if we are currently in the last or previous visited zone, enabling efficient navigation and mapping.
By employing these navigation strategies, we aim to guide our robot successfully through the maze, optimizing its movements and achieving the best possible performance in the Rescue Maze Simulation competition.

c. Victim detection

Initially, we implemented a convolutional neural network (CNN) for image labeling to detect victims within the maze. However, we encountered limitations when it came to accurately locating signs, especially in complex scenarios such as multiple signs present in the field of view. In order to overcome these challenges, we made the challenging decision to transition to an object pose estimation real-time system.

Our new approach combines object detection using the Ultralytics YOLOv8 model, which has been trained on a dataset consisting of approximately 3,000 labeled photos. By leveraging rotation, scalar, and vector multiplication techniques, we have significantly improved our sign detection capabilities. This updated methodology provides more precise and reliable identification of both victims and hazmats present within the maze environment.

d. Mapping

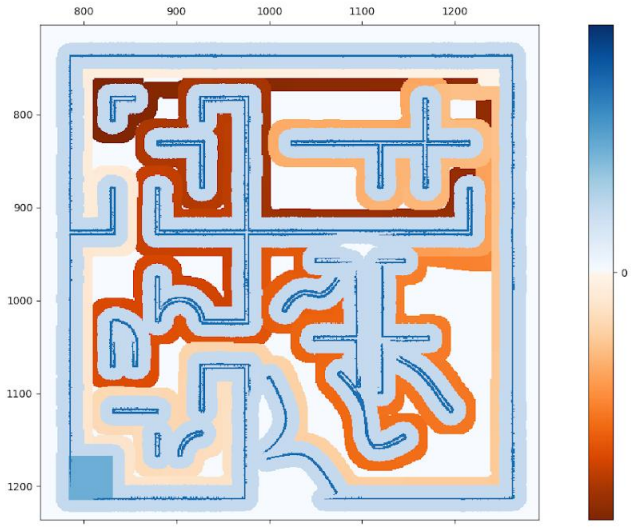
Our team excels in the area of mapping, utilizing LiDAR technology to construct a comprehensive map of the maze. Our mapping solution employs a nonprobabilistic approach, utilizing a grid-based system to differentiate between empty spaces and walls. This allows us to accurately represent the maze environment.

To achieve this, we convert the LiDAR data, which consists of a list of distances measured at equally spaced angles, into a polar representation. This data provides us with a walkable distance in every direction. Subsequently, we transform this polar representation into a Cartesian grid, which is then integrated into the occupancy grid for global mapping.

The conversion process from polar to Cartesian grid involved careful considerations and reflections. We experimented with various approaches, including treating each ray as a vector and rotating it. However, we eventually settled on using an array of indexes as the theoretically fastest method. Currently, we are addressing an issue called extrapolation and working towards resolving it before the competition.

We have gone beyond the minimum requirements for mapping by employing a higher sampling density. Through a down-sampling process, we analyze small sections of the high-resolution grid, checking for wall hits in each area. This enables us to create a bonus map that meets the Erebus requirements, showcasing our commitment to comprehensive and accurate mapping.

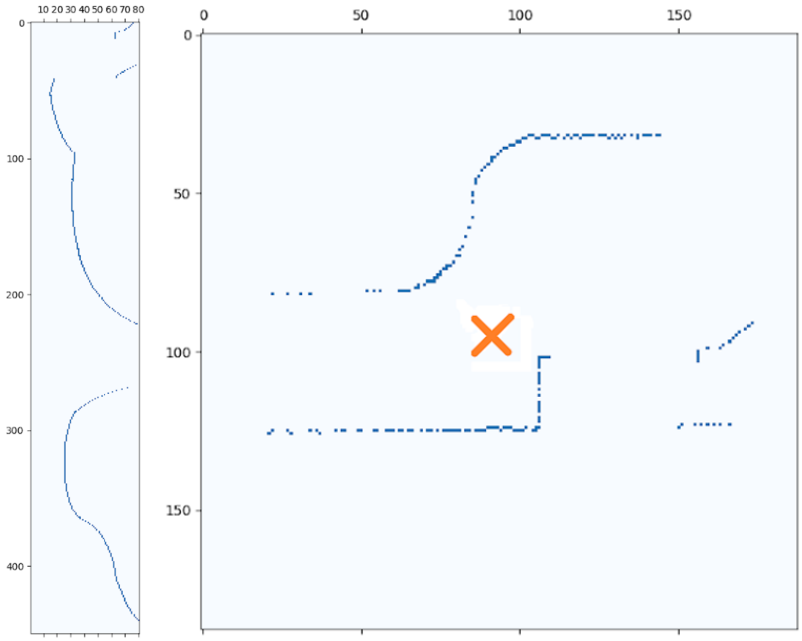
Figure 1: Mapping with Dilation and Visited Regions



This figure illustrates the mapping part of our system, showcasing the inclusion of dilation and visited regions. The map is represented with clear visual distinctions between different areas. Dilation is applied to expand the

occupied space, providing a more accurate representation of the walls and obstacles in the maze. Additionally, the visited regions are marked, indicating the areas already explored by the robot. This information helps in planning future movements and optimizing navigation within the maze.

Figure 2: Polar Grid and Corresponding Cartesian Grid



This figure showcases the polar grid on the left and its corresponding Cartesian grid on the right. The polar grid is a representation of the LiDAR data in a polar coordinate system, capturing the distances measured at equally spaced angles. The Cartesian grid is derived from the polar grid, converting the data into a Cartesian coordinate system. This transformation allows for a more intuitive and structured representation of the environment, enabling efficient mapping and navigation algorithms to be applied.

5. Performance evaluation

We have conducted a thorough evaluation of our robot's performance, highlighting its notable strengths and identifying areas for improvement. While our robot demonstrates several significant advantages in key aspects, we have also encountered certain challenges.

Mapping: Our mapping capabilities exhibit optimal resolution and precision, allowing us to create detailed representations of the maze. However, we currently face difficulty in distinguishing walls from other obstacles within the map. This limitation hinders our ability to accurately differentiate between different types of barriers in the environment.

Navigating: The robot excels at navigating through narrow passages, showcasing its agility and maneuverability. However, we have identified a challenge in maintaining a consistent distance from walls, which affects the efficiency of our sign detection. Ensuring a slight distance from the walls would enhance our ability to detect and recognize signs more effectively.

Victim Recognition: Our robot demonstrates excellent performance in recognizing victims at higher distances. However, we have observed limitations in identifying victims at close proximity. This issue arises from the lack of repositioning to evaluate signs from different angles, affecting our ability to accurately assess signs and detect victims in closer proximity.

Strategy: Unfortunately, due to time constraints, we were unable to fully conclude the necessary aspects of our project related to strategy. As a result, we currently lack a comprehensive strategy that would enable us to

maximize points in larger maps or optimize multiple goal paths once a significant portion of the maze is explored. This represents an area for improvement and future development.

6. Conclusion

In conclusion, we consider it a great honor to participate in such a prestigious competition. We are immensely grateful for the opportunity to showcase our skills, dedication, and passion for computer science, mathematics, and robotics. As the competition progresses, we remain committed to demonstrating our abilities and pushing the boundaries of what is possible in the field. Our team is motivated to learn, adapt, and continually improve our performance, all while embracing the challenges that lie ahead. We are excited to contribute to the advancement of the Rescue Maze Simulation competition and look forward to the experiences and growth it will bring.

References

https://github.com/ultralytics/ultralytics

https://www.sciencedirect.com/science/article/pii/S0921889023000027 (4th point of this paper)